Problem 16.7

Kathy wants to accumulate a sum of money at the end of 10 years to buy a house. In order to accomplish this goal, she can deposit \$80 per month at the beginning of the month for the next ten years or \$81 per month at the end of the month for the next ten years. Calculate the annual effective rate of interest earned by Kathy.

Solution

Let j be the effective monthly interest rate. Since the same accumulated value is reached under either deposit pattern, we equate the accumulated values.

There are

$$10(12) = 120$$

monthly deposits. Deposits of \$80 at the beginning of each month form an annuity due, so their accumulated value is

$$80\ddot{s}_{\overline{120}|}.$$

Deposits of \$81 at the end of each month form an annuity-immediate, so their accumulated value is

$$81s_{\overline{120}|}.$$

Thus,

$$80\ddot{s}_{\overline{120}|} = 81s_{\overline{120}|}.$$

Using

$$\ddot{s}_{\overline{n}|} = (1+j)s_{\overline{n}|},$$

we get

$$80(1+j)s_{\overline{120}|} = 81s_{\overline{120}|}.$$

Canceling $s_{\overline{120}|}$,

$$80(1+j) = 81.$$

Therefore,

$$j = \frac{81}{80} - 1 = 0.0125.$$

The annual effective interest rate is

$$i = (1+j)^{12} - 1.$$

Thus,

$$\begin{aligned} i &= (1.0125)^{12} - 1 \\ &\approx 0.1607545. \end{aligned}$$

Hence, the annual effective rate of interest is

$$\boxed{16.08\%}.$$